Relations

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Chapter Summary

1. Introduction
2. Relations and their properties
3. Representing relations
4. Closure properties
5. Equivalence relations
6. Partial order relations

Binary relations

A and B are two non-empty sets.

Binary Relation: A binary relation R from A to B is a subset of $A \times B$.

Example: A is the set of students. $A = \{Joe, Sam, Peter\}$
B is the set of courses. $B = \{MC123, SC205, IT131\}$

A possible relation is as follows.

$R = \{(Joe, MC123), (Sam, IT131), (Peter, SC205), (Sam, SC205)\}$

Example: A is the states of India B is the cities in India

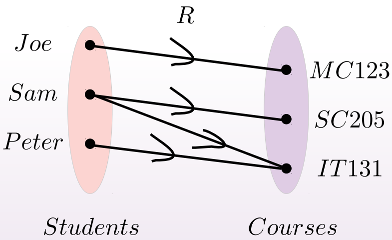
Define a relation R as elements (x, y), y is a city in state x

Then the relation R contains elements as :
$\{(Gujarat, Gandhinagar), (Rajasthan, Jaipur),$
$(West\ Bengal, Kolkata)\}$

Representations of functions

$R = \{(Joe, MC123), (Sam, IT131), (Peter, SC205), (Sam, SC205)\}$

Simple view:



Relations vs functions

- Not all relations are functions
- All functions are relations
- Relations are a generalization of functions

- A function produces a single result for any element in its domain.

  Example: $power(x)$, $sum(x, y)$

- In a relation, it allows multiple mappings between the domain and the codomain.

  Example: students assignment to courses

Representation: using matrices

$R = \{(Joe, MC123), (Sam, IT131), (Peter, SC205), (Sam, SC205)\}$

Tabular way:

	MC123	SC205	IT131
Joe	1		
Sam		1	1
Peter		1	

Matrix representation:

$A = \{a_1, a_2, \ldots, a_n\}$ $\qquad$ $B = \{b_1, b_2, \ldots, b_m\}$

R can be represented as a $(0,1)$-matrix $M_R = [m_{ij}]$, where

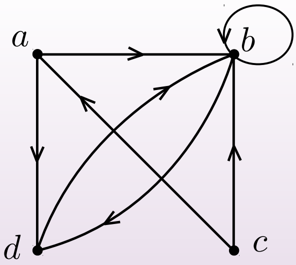
$$m_{ij} = \begin{cases} 1, & \text{if } (a_i, b_j) \in R \\ 0, & \text{if } (a_i, b_j) \notin R \end{cases}$$

Representation: using digraphs

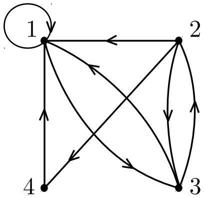
Digraph: Consists of a set V of vertices (or nodes) together with a set E of ordered pairs of elements of V called edges (or arcs).

The vertex a is called the initial vertex of the edge (a, b), and the vertex b is called the terminal vertex of this edge.

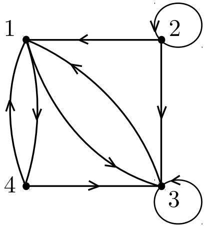
Example: The directed graph with vertices a, b, c, and d, and edges (a, b), (a, d), (b, b), (b, d), (c, a), (c, b), and (d, b) is displayed as follows.



Representation: using digraphs



$R = \{(1,1), (1,3), (2,1), (2,3),$
$(2,4), (3,1), (3,2), (4,1)\}$



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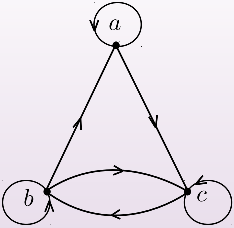
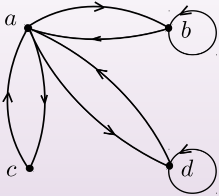




Relation within a set

A relation from a set A to itself.

Example:

A is the set $\{1, 2, 3, 4\}$

$R = \{(a, b) \mid a \text{ divides } b\}$

$R = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)\}$

$R = \{(a, b) \mid a \text{ is the square of } b\}$

$R = \{(1, 1), (4, 2)\}$

Properties of relations: reflexive

Reflexive:
A relation is reflexive if every element is related to itself.

$a \in A$, then $(a, a) \in R$

Example: (i) R is a relation on $\mathbb{R} \to \mathbb{R}$ and is defined by $\{(a, b) \mid a = b\}$
It is reflexive, as any element/number is equal to itself.

(ii) $R = \{(a, b) \mid a \leq b\}$ Yes

(iii) $R = \{(a, b) \mid a < b\}$ No! $(a, a) \notin R$ irreflexive

Symmetric

A relation is symmetric if for every $(a, b) \in R$ then $(b, a) \in R$.

Example: $\{(x, y) \mid x\ is\ a\ friend\ of\ y\}$ YES

$\{(x, y) \mid x\ is\ a\ sibling\ of\ y\}$ YES

$\{(x, y) \mid x\ is\ a\ brother\ of\ y\}$ NO, YES, when x, y are males

Asymmetric: A relation is asymmetric if for every $(a, b) \in R$ then $(b, a) \notin R$.

Example: $\{(x, y) \mid x\ is\ a\ parent\ of\ y\}$

Antisymmetric

A relation is antisymmetric if $(a, b) \in R$ and $(b, a) \in R$, then $a = b$.

Example: (i) $\{(x, y) \mid x \leq y\}$

(ii) $\{(x, y) \mid x = |y|\}$

Not symmetric as $(1, -1) \in R$ but $(-1, 1) \notin R$

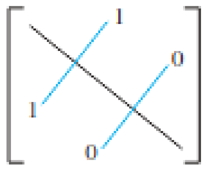
antisymmetric Yes

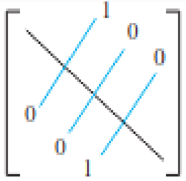
Not asymmetric as $(1, 1) \in R$ but need to prove $(1, 1) \notin R$

Representation: using matrices

Reflexive: $\begin{bmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{bmatrix}$

Symmetric and antisymmetric:





Transitive

A relation is transitive if $(a, b) \in R$ and $(b, c) \in R$, then $(a, b) \in R$.

Example: (i) $\{(x, y) \mid x \; is \; an \; ancestor \; of \; y\}$ YES

(ii) $\{(x, y) \mid x \; is \; a \; parent \; of \; y\}$ NO

(ii) $\{(x, y) \mid x \; is \; a \; sibling \; of \; y\}$ YES

Operations

$A = \{1, 2, 3, 4, 5\}$ $\qquad$ $B = \{a, b, c\}$

$R_1 = \{(1, a), (2, b), (4, c), (4, b)\}$

$R_2 = \{(1, a), (3, b), (5, c), (4, b)\}$

- $R_1 \cup R_2 = \{(1, a), (2, b), (4, c), (4, b), (3, b), (5, c)\}$
- $R_1 \cap R_2 = \{(1, a), (4, b)\}$
- $R_1 \setminus R_2 = \{(2, b), (4, c)\}$

Composition

R is a relation from A to B S is a relation from B to C

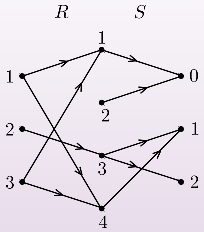
then the composition of R and S i.e., $S \circ R$ is
$\{(a, c) \mid a \in A, b \in B, c \in C \text{ such that } (a, b) \in R, (b, c) \in S\}$

Example: $A = \{1, 2, 3\}$ $B = \{1, 2, 3, 4\}$ $C = \{0, 1, 2\}$

$R = \{(1, 1), (1, 4), (2, 3), (3, 1), (3, 4)\}$

$S = \{(1, 0), (3, 1), (3, 2), (4, 1)\}$

$S \circ R = \{(1, 0), (1, 1), (2, 1), (2, 2), (3, 0), (3, 1)\}$



$1 \to 1 \to 0 \quad (1, 0)$
$1 \to 4 \to 1 \quad (1, 1)$
$2 \to 3 \to 1 \quad (2, 1)$
$2 \to 3 \to 2 \quad (2, 2)$
$3 \to 1 \to 0 \quad (3, 0)$
$3 \to 4 \to 1 \quad (3, 1)$

Composition on a set

$R^n = R^{n-1} \circ R = R \circ R \circ R \circ \ldots \circ R$

$R = \{(1,1),(2,1),(3,2),(4,3)\}$

$R^2 = R \circ R = \{(1,1),(2,1),(3,1),(4,2)\}$

$R^3 = R^2 \circ R = \{(1,1),(2,1),(3,1),(4,1)\}$

$R^4 = R^3 \circ R = \{(1,1),(2,1),(3,1),(4,1)\}$

For $n > 3$, $R^n = R^3$

Matrix representation of operations

Example: Find the matrix representing the relations $S \cup R$, and $S \cap R$ where the matrices representing R and S are

$$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \qquad M_S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

Solution: The matrix for $S \cup R$ is:

$$M_{S \cup R} = M_S \vee M_R = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

$$M_{S \cap R} = M_S \wedge M_R = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Matrix representation of composition

- R: relation from A to B S: relation from B to C.
- Let A, B, and C have m, n, and p elements, respectively.
- Let the $0-1$-matrices for $S \circ R$, R, and S be $M_{S \circ R} = [t_{ij}]$, $M_R = [r_{ij}]$, and $M_S = [s_{ij}]$, respectively (these matrices have sizes $m \times p$, $m \times n$, and $n \times p$, respectively).
- The ordered pair (a_i, c_j) belongs to $S \circ R$ if and only if there is an element b_k such that (a_i, b_k) belongs to R and (b_k, c_j) belongs to S.
- It follows that $t_{ij} = 1$ if and only if $r_{ik} = s_{kj} = 1$ for some k.
- The Boolean product,
  $M_{S \circ R} = M_R \odot M_S$. $t_{ij} = \vee_{k=1}^{n}(r_{ik} \wedge s_{kj})$
- $M_{R^n} = M_R^{[n]}$, the Boolean n-th power. $[M_{R \circ R} = M_{R^2} = M_R^{[2]}]$

Matrix representation of composition

Example: Find the matrix representing the relations $S \circ R$, where the matrices representing R and S are

$$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \qquad M_S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

Solution: The matrix for $S \circ R$ is:

$$M_{S \circ R} = M_R \odot M_S = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{R \circ R} = M_{R^2} = M_R \odot M_R = M_R^{[2]} \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Closure

Let R be a relation on a set A. The relation R may or may not have some property P such as reflexivity, symmetry, or transitivity.

If there is a relation S

- with property P,
- containing R,
- and such that S is a subset of every relation with property P containing R,

then S is called the closure of R with respect to P.

Reflexive closure

A relation R on a set A is called reflexive if $(a, a) \in R$ for every element $a \in A$.

Example: Let A be the set $\{1, 2, 3, 4\}$ and R be the relation $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\}$.

$$M_R = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

NOT reflexive

Reflexive closure: Let R be a relation on a set A. The reflexive closure of R is $R \cup \Delta$, where $\Delta = \{(a, a) \mid a \in A\}$ is called the diagonal relation on A.

Symmetric closure

A relation R on a set A is called symmetric if $(a, b) \in R$ implies that $(b, a) \in R$ for all element $a, b \in A$.

Example: Let A be the set $\{1, 2, 3, 4\}$ and R be the relation $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\}$.

$$M_R = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

NOT symmetric

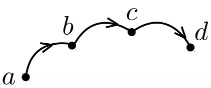
Symmetric closure: Let R be a relation on a set A. The symmetric closure of R is $R \cup R^{-1}$, where $R^{-1} = \{(b, a) \mid (a, b) \in R\}$ is called the inverse relation of R.

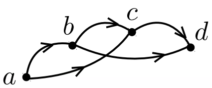
Transitive cosure

Transitive relation: If both (a,b) and (b,c) are in the relation, then add (a,c) to make the relation transitive.

But that does not work. Why?

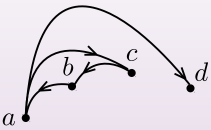
Example 1:

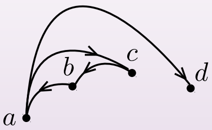




The relation (a,d) is not captured.

Example 2:





The relation (c,d) is not captured.

Path

- A graph G is set of V vertices and E edges (edge is an ordered pair (a, b) of vertices)
- The path from a to b in G is a sequence of edges (x_i, x_{i+1}), denoted by $x_0, x_1, x_2, \ldots, x_{n-1}, x_n$ (length n) where $x_0 = a$ and $x_n = b$.
- A circuit or cycle is a path of length $n \geq 1$ that begins and ends at the same vertex.

Transitive closure

Theorem: Let R be a relation on a set A. There is a path of length n, where n is a positive integer, from a to b if and only if $(a, b) \in R^n$.

Connectivity relation

Connectivity relation: Let R be a relation on a set A. The connectivity relation R^* consists of the pairs (a, b) such that there is a path of length at least 1 from a to b in R.

R^n consists of the pairs (a, b) such that there is a path of length n from a to b

$$R^* = \bigcup_{n-1}^{\infty} R^n$$

Transitive closure

Example: Let R be the relation on the set of all people in the world that contains (a, b) if a has met b. What is R^n, where n is a positive integer greater than one? What is R^*?

Solution:

- The relation R^2 contains (a, b) if there is a person c such that $(a, c) \in R$ and $(c, b) \in R$, that is, if there is a person c such that a has met c and c has met b.
- R^n consists of those pairs (a, b) such that there are people $x_1, x_2, \ldots, x_{n-1}$ such that a has met x_1, x_1 has met x_2, $\ldots$, and x_{n-1} has met b.
- The relation R^* contains (a, b) if there is a sequence of people, starting with a and ending with b, such that each person in the sequence has met the next person in the sequence.

Transitive closure

Theorem: The transitive closure of a relation R equals to R^*.

Proof sketch:

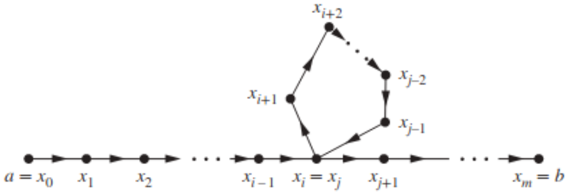
- R^* contains R by definition
- R^* is transitive —- just follow the paths
- Any transitive relation S that contains R must also contain R^*

Transitive closure

Theorem: Let A be a set with n elements, and let R be a relation on A. If there is a path of length at least one in R from a to b, then there is such a path with length not exceeding n.

Moreover, when $a \neq b$, if there is a path of length at least 1 in R from a to b, then there is such a path with a length not exceeding $n - 1$.

Proof sketch:



Proof

- Suppose there is a path from a to b in R.
- Let m be the length of the shortest such path.
- Suppose that $x_0, x_1, x_2, \ldots, x_{m-1}, x_m$, where $x_0 = a$ and $x_m = b$, is such a path.
- Suppose that $a = b$ and that $m > n$, so that $m \geq n + 1$.
- By the pigeonhole principle, because there are n vertices in A, among the m vertices $x_0, x_1, x_2, \ldots, x_{m-1}$, at least two are equal.
- Suppose that $x_i = x_j$ with $0 \leq i < j \leq m - 1$.
- Then the path contains a circuit from x_i to itself.
- This circuit can be deleted from the path from a to b, leaving a path, namely, $x_0, x_1, \ldots, x_i, x_{j+1}, \ldots, x_{m-1}, x_m$, from a to b of shorter length.

  The case where $a \neq b$ is left as an exercise for the reader.

Transitive closure

Theorem: M_R be the zero–one matrix of the relation R on a set with n elements. Then the zero–one matrix of the transitive closure R^* is

$$M_{R^*} = M_R \vee M_R^{[2]} \vee M_R^{[3]} \vee \ldots \vee M_R^{[n]}.$$

Example: Find the zero–one matrix of the transitive closure of the relation R where

$$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

Solution: $M_{R^*} = M_R \vee M_R^{[2]} \vee M_R^{[3]}$

$$M_{R^*} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \vee \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \vee \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

n-ary relation

n-ary relation: Let $A_1, A_2, \ldots, A_n$ be n sets.

Then, a n-ary relation is a subset of $A_1 \times A_2 \times \ldots \times A_n$

Example: R is a relation on $\mathbb{R} \times \mathbb{N} \times \mathbb{Z}$ then

- $(0.5, 3, -2) \in R$
- $(0.5, -5, 2) \notin R$
- $(0.6, 5, 2.5) \notin R$

Application: Relational data model

Student record

s_id	s_name	course	grade
20230001	Sam	SC205	AB
20230005	Mike	SC205	AA
20230007	David	IT102	BB

(20230001, Sam, SC205, AB)

Equivalence relation

Equivalence relation: An equivalence relation on a set S is a relation that is reflexive, symmetric, and transitive.

Example: Let $A = \{1, 2, 3, 4\}$

R is a binary relation defined on A as
$R = \{(1, 1), (2, 2), (2, 4), (3, 3), (4, 2), (4, 4)\}$

Is R an equivalence relation?

R is reflexive, symmetric, and transitive.

Equivalence relation: example

Congruences: Let a and b be integers, and let n be a positive integer. Then a is b congruent to modulo n denoted by $a \equiv b \ (mod \ n)$
This means $n|(a-b)$
i.e., there exists an integer such that $(a-b) = k \cdot n$
Example:

- Is $18 \equiv 4 \ (mod \ 7)$?
  $18 - 4 = 14 = 7 \times 2$
  Hence $7|(18-4)$

Equivalence relation: example

R is a relation congruence modulo n on $\mathbb{Z}$, i.e., aRb is defines as $a \equiv b \ (mod \ n)$, for all $a, b \in \mathbb{Z}$. Prove that R is an equivalence relation.

Solution:

- R is reflexive.
  - $a \equiv a \ (mod \ n)$ for all $a \in \mathbb{Z}$
  - $a - a = 0 = n \times 0$
  - $n|(a - a)$
- R is symmetric.
  - if aRb then bRa that is, if $a \equiv b \ (mod \ n)$ then $b \equiv a \ (mod \ n)$
  - $aRb \implies a \equiv b \ (mod \ n) \implies n|(a - b) \implies a - b = n \times k$
  - $\implies b - a = n \times (-k) \implies n|(b - a) \implies b \equiv a \ (mod \ n) \implies bRa$
- R is transitive.
  - if aRb and bRc then aRc that is, if $a \equiv b \ (mod \ n)$ and $b \equiv c \ (mod \ n)$ then $a \equiv c \ (mod \ n)$
  - $n|(a - b)$ and $n|(b - c)$ $\quad (a - b) = n \times k$ and $(b - c) = n \times \ell$
  - $(a - b) + (b - c) = n \times k + n \times \ell \quad (a - c) = n \times k'$ where $k' = k + \ell$
  - $n|(a - c) \implies a \equiv c \ (mod \ n) \implies aRc$

Equivalence relation: example

Let n be a positive integer and S a set of strings. Suppose that R_n is the relation on S such that $(s,t) \in R_n$ if and only if $s = t$, or both s and t have at least n characters and the first n characters of s and t are the same.

For example, let $n = 3$ and let S be the set of all bit strings. Then $(s,t) \in R_3$ either when $s = t$ or both s and t are bit strings of length 3 or more that begin with the same three bits.

For instance, $(01, 01) \in R_3$ and $(00111, 00101) \in R_3$, but $(01, 010) \notin R_3$ and $(01011, 01110) \notin R_3$.

Solution: Classwork

Equivalence relation: example

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Solution: Classwork

Equivalence classes

Equivalence classes: Let R be an equivalence relation on a set A and a be an element of A. Let $[a]_R = \{b \mid aRb\}$ be the equivalence class of a w.r.t. R, i.e., all elements of A that are R-equivalent to a.

If $b \in [a]_R$ then b is called a **representative** of the equivalence class.

Any member of the class can be a representative. (no special rule)

Example: What are the equivalence classes of 0, 1, 2 for congruence modulo 3 ?

Solution: The equivalence class of 0 contains all integers a such that $a \equiv 0 \ (mod\, 3)$. The integers in this class are those divisible by 3.

$[0] = \{\dots, -6, -3, 0, 3, 6, \dots\}$.

$[1] = \{\dots, -5, -2, 1, 4, 7, \dots\}$.

$[2] = \{\dots, -4, -1, 2, 5, 8, \dots\}$.

Equivalence classes

Example: What is the equivalence class of the string 0111 with respect to the equivalence relation R_3 from Example defined earlier on the set of all bit strings? (Recall that $(s, t) \in R_3$ if and only if s and t are bit strings with $s = t$ or s and t are strings of at least three bits that start with the same three bits.)

Solution: The bit strings equivalent to 0111 are the bit strings with at least three bits that begin with 011.

These are the bit strings $011, 0110, 0111, 01100, 01101, 01110, 01111,$ and so on.

Consequently,
$[011]_{R_3} = \{011, 0110, 0111, 01100, 01101, 01110, 01111, \ldots\}$.

Equivalence classes and partitions

Theorem: If R is an equivalence on A, then the equivalence classes of R form a partition of A.

Conversely, given a partition $\{A_i \mid i \in I\}$ of A there exists an equivalence relation R that has exactly the sets $A_i, i \in I$, as its equivalence classes.

How to construct equivalence relation from partition:

Example: List the ordered pairs in the equivalence relation R produced by the partition $A_1 = \{1, 2, 3\}$, $A_2 = \{4, 5\}$, and $A_3 = \{6\}$ of $S = \{1, 2, 3, 4, 5, 6\}$

Solution: The subsets of S in the partition are the equivalence classes of R. The pair $(a, b) \in R$ if and only if a and b are in the same subset of the S in the partition. The pairs $(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2),$ and $(3, 3)$ belong to R because $A_1 = \{1, 2, 3\}$ is an equivalence class; the pairs $(4, 4), (4, 5), (5, 4), (5, 5)$ belong to R because $A_2 = \{4, 5\}$ is an equivalence class; and finally the pair $(6, 6)$ belongs to R because $\{6\}$ is an equivalence class. No pair other than those listed belongs to R.

Partial orders

Partial orders: A relation R on a set S is called a partial order iff it is reflexive, antisymmetric and transitive.

If R is a partial order, we call (S, R) a partially ordered set, or poset.

Example: $\leq$ is a partial order, but $<$ is not (since it is not reflexive).

Example: Let $a|b$ denote the fact that a divides b. Formally: $\exists k \in \mathbb{Z}, b = ak$. Show that the relation $|$ is a partial order, i.e., $(Z^+, |)$ is a poset.

Example: Set inclusion $\subseteq$ is partial order, i.e., $(2^A, \subseteq)$ is a poset.

Solution:

- Reflexive, $S \subseteq S$
- Symmetric, if $S \subseteq T$ and $T \subseteq S \implies S = T$
- Transitive, if $S \subseteq T$ and $T \subseteq W$, then $S \subseteq W$

Comparability and Total Orders

Two elements a and b of a poset (S, R) are called **comparable** iff aRb or bRa hold.

Otherwise, they are called **incomparable**.

Example: $(\mathbb{Z}^+, |)$ is a poset.

Question: Is 3 and 12 comparable? Ans: Yes

Is 7 and 9 comparable? Ans: No

Example: $(S, \subseteq)$ be a poset. $S = \{x, y, z\}$

Question: Is $\{x, y\}$ and $\{x, z\}$ comparable? Ans: No

Is $\{x\}$ and $\{x, z\}$ comparable? Ans: Yes

Totally ordered set

Totally ordered set: If (S, R) is a poset where every two elements are comparable, then S is called a totally ordered or linearly ordered set, and the relation R is called a total order or linear order.

A totally ordered set is also called a **chain**.

Given a poset (S, R) and $S' \subseteq S$ a subset in which all elements are pairwise incomparable. Then S' is called an **antichain**.

Self study

- Hasse diagram
- Least upper bound
- Greatest lower bound
- Lattice